

CoGenASD: A tool for the co-design and generation of cross-platform applications for people with Autism spectrum disorder

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ABSTRACT

Current ASD-focused app development faces key limitations, such as high technical barriers for non-experts, limited personalization, and scarce involvement of therapists, families and educators in the design process. This paper presents **CoGenASD**, a framework that integrates co-design principles with a Model-Driven Development (MDD) approach to support the semi-automatic generation of cross-platform applications for individuals with ASD. The tool enables multidisciplinary teams (therapists, families and educators) to collaboratively define and model participant profiles, activities, interaction modes and content, supporting the semi-automatic generation of cross-platform, accessible and tailored applications. CoGenASD lowers technical barriers, promotes inclusive design practices, and accelerates the development of support tools. Its potential impact includes increasing application effectiveness, fostering stakeholder engagement, and enabling new research on customizable interventions for neurodiverse populations.

0 Metadata

Nr	Code metadata description	Metadata
C1	Current code version	1.0.0.
C2	Permanent link to code/repository used for this code version	https://github.com/ChicoUclm/cogenasd
C3	Permanent link to reproducible capsule	https://codeocean.com/capsule/0652889/tree
C4	Legal code license	MIT License
C5	Code versioning system used	Git
C6	Software code languages, tools and services used	Eclipse, Emfatic, Epsilon, Epsilon Validation Language, Epsilon Generation Language, Epsilon Object Language, Eclipse Communication Framework, Java, HTML, JavaScript, CSS
C7	Compilation requirements, operating environments and dependencies	Eclipse Epsilon
C8	If available, link to developer documentation/manual	
C9	Support email for questions	Yoel.Arroyo@uclm.es

1. Motivation and significance

Autism Spectrum Disorder (ASD) encompasses a range of neuro-developmental conditions that affect communication, social interaction and behaviour [1]. ASD is a highly heterogeneous condition, meaning that each individual exhibits unique manifestations and needs, requiring personalized support and care [2]. In this context, **Augmentative and Alternative Communication (AAC)** systems,¹ predominantly grounded in visual resources (typically pictograms²) [3,4], have proven to be highly effective in fostering anticipation of situations and the expression of emotions, both of which constitute essential abilities for adaptive functioning and social integration [5].

Although **digital technologies** offer promising avenues to support individuals with ASD, developing customized applications that effectively address their unique needs remains a complex challenge [6].

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¹ *Augmentative and Alternative Communication Systems (AACS)* are forms of expression other than spoken language that aim to increase the level of expression (augmentative) and/or compensate (alternative) for the communication difficulties that some people have in this area.

² A *pictogram* is a simplified graphical representation of an object, concept or action, designed to convey meaning quickly and unambiguously. In the context of ASD, pictograms are widely used as visual supports within Augmentative and Alternative Communication (AAC) systems to facilitate understanding, anticipation of routines, and expression of needs or emotions.

Existing solutions often require high technical expertise, lack adaptability across devices, and rarely involve end-users or stakeholders in the design process. For example, many communication or planning applications provide resources or routines in fixed formats that either cannot be customized or require complex configurations to adapt them to individual preferences or cognitive levels [7]. This forces users and caregivers to combine multiple tools or rely heavily on professional support to configure them to meet individual needs, which can lead to rejection of this type of technological solution. In response, recent advances emphasize participatory and structured methodologies that actively involve end-users, therapists and caregivers throughout the process [8, 9], moving away from traditional designer-driven approaches toward frameworks based on co-design principles, iterative prototyping and recommendations tailored to ASD needs [10]. To address these limitations, CoGenASD was developed as a semi-automatic software tool that leverages Model-Driven Development (MDD) to facilitate co-design, support user requirements specification into high-level models, validation of design guidelines and the semi-automatic generation of adaptive, cross-platform applications tailored to individuals with ASD. MDD [11] is a software development methodology that uses models, often visual, as the primary artifact to create software. Instead of manually writing all code, developers use these models to specify a system's structure and behaviour, which are then transformed into executable code through automated processes. This approach focuses on abstraction to simplify complexity, automation to accelerate development, and improves collaboration by providing a common, understandable blueprint for user needs. The relevance of CoGenASD lies in combining co-design with MDD to empower both developers and non-technical stakeholders—such as therapists, educators and families—in collaboratively creating personalized, accessible applications across devices. Thus, this tool addresses a significant gap in the **assistive technology** field by lowering the barrier to developing customized solutions while preserving user-centred design principles.

CoGenASD has been developed as an Eclipse-based plugin using the Epsilon³ framework to implement a **model-driven approach** that supports **participatory design methodologies** [9,12]. Through a **domain-specific modelling language (DSML)**, therapists and caregivers can collaboratively define the structure and behaviour of pictogram-based applications [3,4] for communication, planning or emotion recognition, using an intuitive visual environment accessible to non-technical stakeholders [13]. The plugin reduces manual coding and enables the generation of adaptive applications for multiple devices (smartphones, tablets, large displays), thus combining technical automation with participatory design to deliver efficient, inclusive and customizable digital supports tailored to individuals with ASD.

2. Software description

The CoGenASD framework is comprised of three main software components: (1) the Graphical User Interface (GUI) modelling component, (2) the co-design and collaboration component, and (3) the export and deployment component.

Fig. 1 presents, from a general perspective, the architecture that underpins CoGenASD and illustrates the relationships between these core components. Thus, the **GUI Component** provides the ASD graphical editor, built on EMF/GEF/GMF, which automatically transforms models using predefined templates. The **Co-Design and Collaboration Component** integrate the SpacEclipse [14] plugin with Eclipse Communication Framework (ECF) remote services, allowing real-time synchronization, messaging and shared editing across multiple users. Finally, the **Export and Deployment Component** generate a multi-device ASD application (using HTML, CSS, JavaScript and images) so diagrams can be deployed on web, mobile and desktop platforms.

2.1. GUI modelling component

The GUI modelling component (Fig. 1.a) allows therapists and ASD specialists to design the content and structure of activities aimed at users with ASD, using an **Eclipse graphical editor**. This tool incorporates predefined *templates* and *reusable components* that speed up the design process. The graphic editor includes a *palette* (Fig. 2) with icons representing the main elements of the **DSML** proposed for specifying activities aimed at improving the social skills of people with ASD [13]. So far, the editor supports the modelling of the following two types of activities (Fig. 3):

1. The **planning and anticipation of daily routines** activities (Fig. 3) are specified using a generic graphical notation comprising *nodes*, *links* and *annotations* [13]. *Nodes* depict elements such as people, places or actions, while *links* define the temporal or logical sequence among them. Both *nodes* and *links* can be annotated with supportive information to enhance understanding and engagement. Supported annotations include:
 - *Social recommendations to guide interpersonal communication* (Fig. 3a). These individualized narratives help people with ASD comprehend social cues and improve interaction skills [15–17].
 - *Augmentative and alternative communication (AAC) tools*, such as the AAC notebook, can be referenced at specific steps where their use may be required. Recent studies highlight the benefits of personalized AAC approaches for improved communication and cognitive engagement in users with ASD [18,19].
 - *Temporal information*, like *waiting times* or *deadlines* (Fig. 3b), which are essential for helping individuals with ASD anticipate changes and manage anxiety [20].
 - *Rewards and incentives*, emphasizing the role of positive reinforcement (well-supported by ASD intervention literature) to boost motivation and engagement [21].
 - *Calming activities or objects* (Fig. 3c), such as toys, with sensory qualities, support self-regulation during stress and help manage anxiety, particularly in waiting situations [22].
2. For **emotion recognition** activities (Fig. 3), the diagrams incorporate *nodes* representing a target emotion (e.g., a facial expression or social scene) (Fig. 3d) and *selectable response options* (Fig. 3e). *Links* between these nodes specify correct or incorrect answers and allow the inclusion of positive or negative feedback.

The activities designed comply with universal design principles and accessibility standards (e.g., WCAG, AAC) using multimodal cues and predictable visual patterns. It also takes into account different sensory and communicative preferences, reduces cognitive load, and complies with specific guidelines for designing applications for users with ASD [23].

The CoGenASD graphical editor empowers practitioners (therapists or ASD specialists) not only to design personalized ASD activity diagrams but also to seamlessly initiate the generation of cross-platform applications for individuals with ASD. This capability is enabled thanks to the use of the Eclipse Epsilon environment, a powerful platform for MDD that automates key model-based engineering tasks [11]. Table 1 lists the technologies used in this first component of the framework.

Once a practitioner completes an ASD activity diagram, the system can automatically trigger the generation of a ready-to-deploy application. This transformation is driven by a set of predefined templates created using the Epsilon Generation Language (EGL),⁴ specifically contained within the *Templates.egl* file and orchestrated by the *Rules.egx* file (Fig. 1a). These components, though transparent to the practitioner, enable the high-level visual specification (diagram) to be translated into

³ <https://eclipse.dev/epsilon/>

⁴ <https://www.eclipse.org/epsilon/doc/egl/>

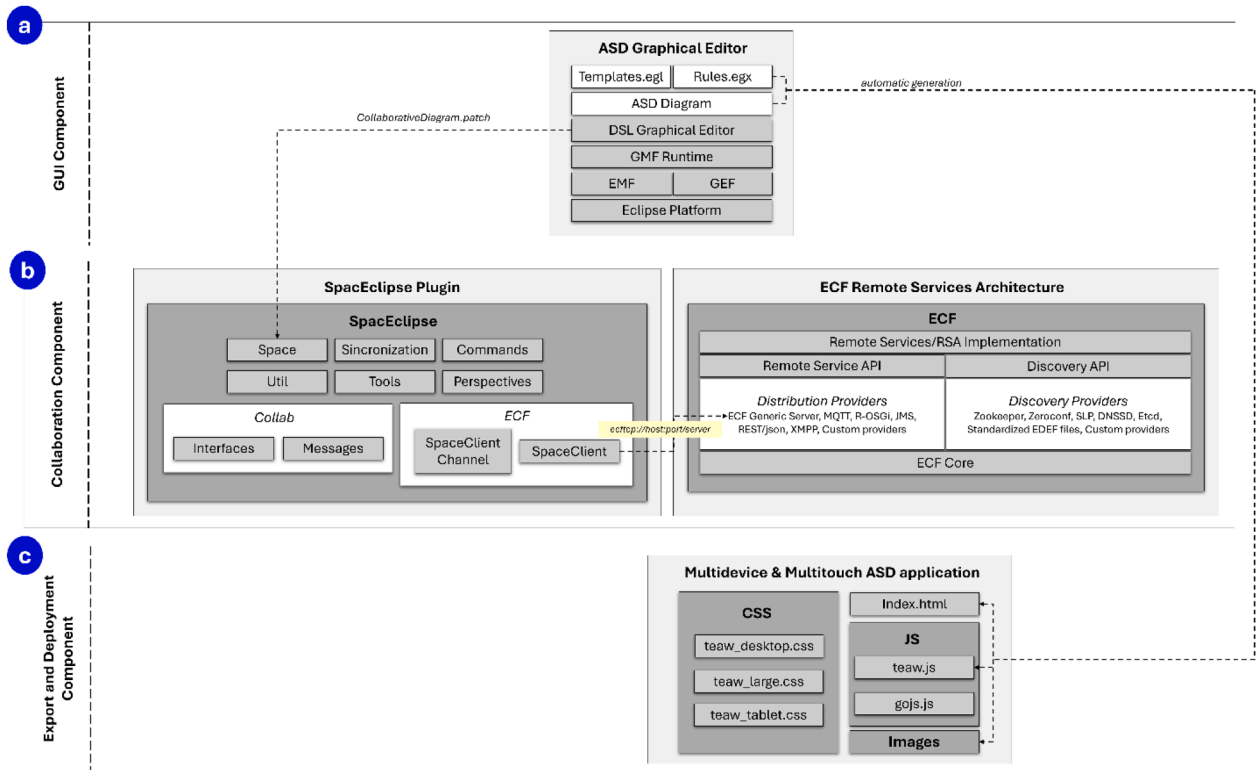


Fig. 1. Overview and key components of the CoGenASD architecture.

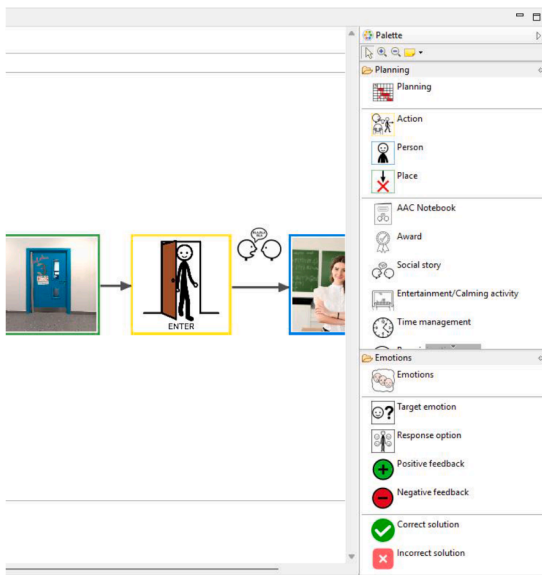


Fig. 2. CoGenASD palette options available, including categories for Planning, AAC Notebook and Emotions.

a functional, cross-platform ASD application.

2.2. Co-design and collaboration component

CoGenASD supports both *mono-user* and *collaborative* modes. Collaborative mode is enabled through the co-design and collaboration component (Fig. 1b), which allows support persons for individuals with ASD (practitioners and family members or caregivers) to jointly participate in the design process, sharing insights and decisions in real time. During this collaborative customization phase, stakeholders instantiate

the activity templates (specified with the proposed DSML) and import images from external sources, allowing activities to be tailored to the preferences and needs of ASD users.

This first component (GUI modelling component) can support collaborative use (Fig. 1b) through the integration of SpacEclipse [14], a plugin for Eclipse that provides model-driven support for the development of collaborative tools. Table 2 enumerates the technologies and languages involved in this second component of the framework.

SpacEclipse provides abstractions and reusable components to manage aspects such as coordination, communication and *awareness*,⁵ enabling developers to create collaborative modelling tools and applications in a systematic and efficient way. This functionality relies on the Eclipse Communication Framework (ECF) (Fig. 1b), an open-source middleware for distributed applications that provides messaging, presence, and shared service capabilities, serving as the communication backbone of SpacEclipse.

When the CoGenASD graphical editor and SpacEclipse are present in the same Eclipse workspace, their integration can be activated using the *CollaborativeDiagram.patch* (Fig. 1b). This patch modifies the main SpacEclipse class to specifically enable support for collaborative design of activity specification diagrams. Integration is technically facilitated through two main internal components: the *Space* Java class (Fig. 1b), which adapts any generic DSL graphical editor, such as the activity diagram editor, to SpacEclipse's collaborative environment; and the *SpaceClient* class (Fig. 1b), which connects SpacEclipse with an ECF Generic Server, allowing for real-time multi-user interaction among practitioners and support providers for individuals for ASD.

The collaborative and distributed editing feature supported by

⁵ *Awareness* refers to understanding the activities of others to support one's own work. It is commonly conveyed through mechanisms such as participant color-coding, telepointers, presence indicators or notifications of changes in the shared workspace. Such awareness mechanisms enhance coordination and improve the efficiency of collaborative work.

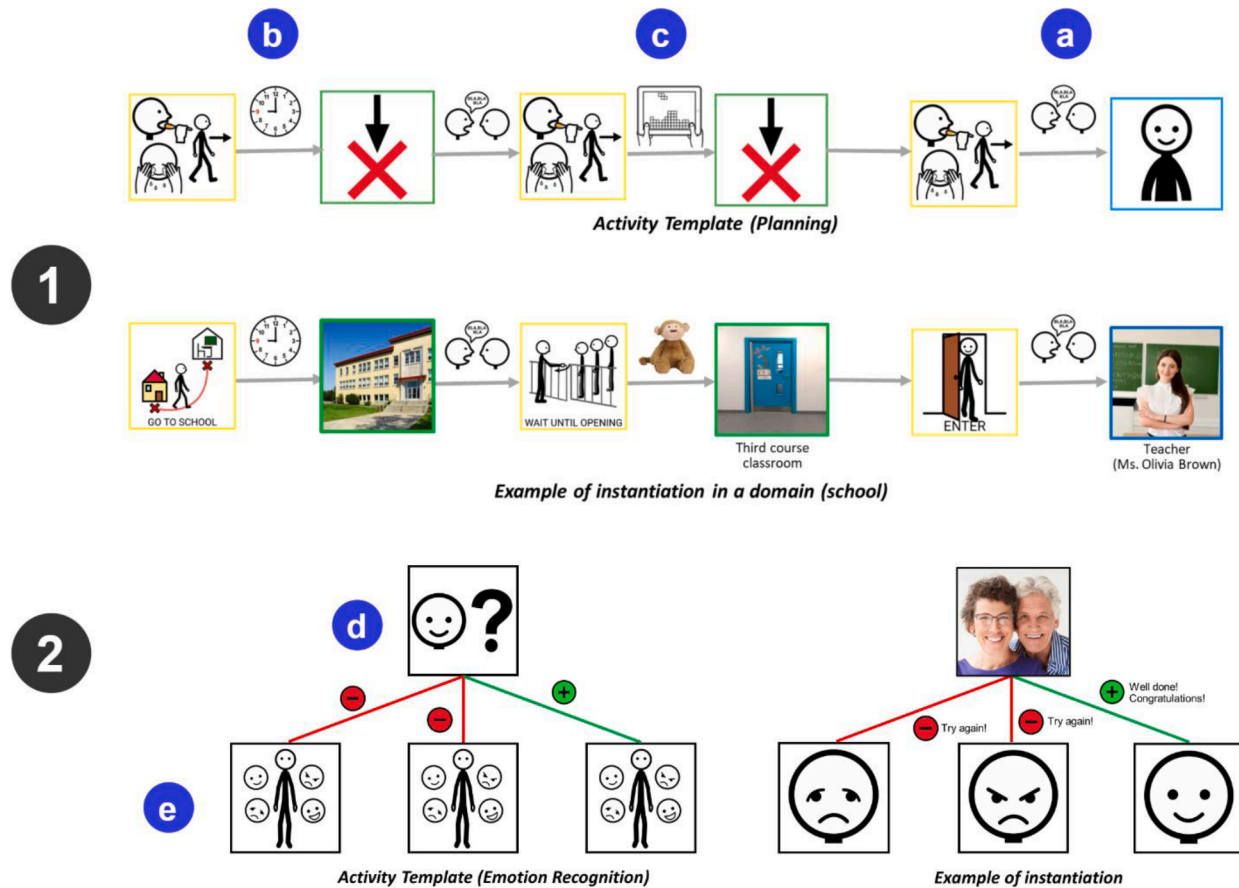


Fig. 3. The CoGenASD graphical editor supports visual specification diagrams for: (1) planning and anticipation activities, and (2) emotion recognition. The figure shows examples of an activity template and a possible instantiation of that template.

Table 1

Summary of technologies and languages used in the GUI modelling component.

Technology	Definition	Use in CoGenASD
Eclipse Epsilon	Model management framework providing a set of domain-specific languages (DSLs) to support model-driven development (MDD) processes	Design, validation and generation of the CoGenASD graphical editor
EGL (Epsilon Generation Language)	Language for generating text or code from models	Enables the automatic generation of cross-platform applications from the specifications (diagrams) of the ASD activities (.egl file)

Table 2

Summary of technologies and languages used in the collaboration component.

Technology	Definition	Use in CoGenASD
SpacEclipse	Plug-in enabling collaboration features (chat, session panel, turn-taking panel) in Eclipse	Initiates collaborative sessions in the CoGenASD graphical editor
EVL (Epsilon Validation Language)	DSL for defining validation constraints	Enforces ASD diagrams validity usability and therapeutic rules
ECF (Eclipse Communication Framework)	Framework for real-time and asynchronous communication in Eclipse	Enables real-time collaboration in the CoGenASD graphical editor

CoGenASD is especially valuable when remote participation by family members or caregivers is necessary. SpacEclipse enhances the graphical editor with a suite of collaborative tools (Fig. 4), including a *chat*, a *turn-taking panel* and a *session panel*. It also improves user awareness by employing *color-coded identifiers*, *telepointers*, *semaphores* to manage editing shifts and *sound notifications* to indicate GUI changes in real time. These collaborative mechanisms support group decision-making and *participatory design* [8,9] sessions, enabling stakeholders to jointly discuss, negotiate, refine and validate activity designs in real time. In practice, this environment also facilitates focus-group-style sessions, where therapists, educators and families can collaboratively contribute requirements and feedback during the design process [24].

A notable feature of CoGenASD is the **validation of activity designs** in both individual and collaborative editing. Using Epsilon Validation Language (EVL),⁶ structural and syntactic constraints can be specified, such as, for ensuring all nodes in a plan are connected or limiting the number of pictograms (Fig. 5a). The editor generates *warnings* or *errors* for unmet constraints, which can be corrected manually or automatically (Fig. 5b). In addition, **validations related to usability and accessibility** guidelines have been incorporated based on a UX checklist for evaluating applications aimed at ASD users [23]. For instance, the guideline “the system provides rewards to the user for positive actions or good performance” has been implemented in EVL, requiring the inclusion of an award or recompense node in planning activities (Fig. 5c). By encoding these guidelines as EVL-based validation rules, which are automatically evaluated, CoGenASD integrates them directly into the modelling workflow, guiding stakeholders toward accessible,

⁶ <https://www.eclipse.org/epsilon/doc/evl/>

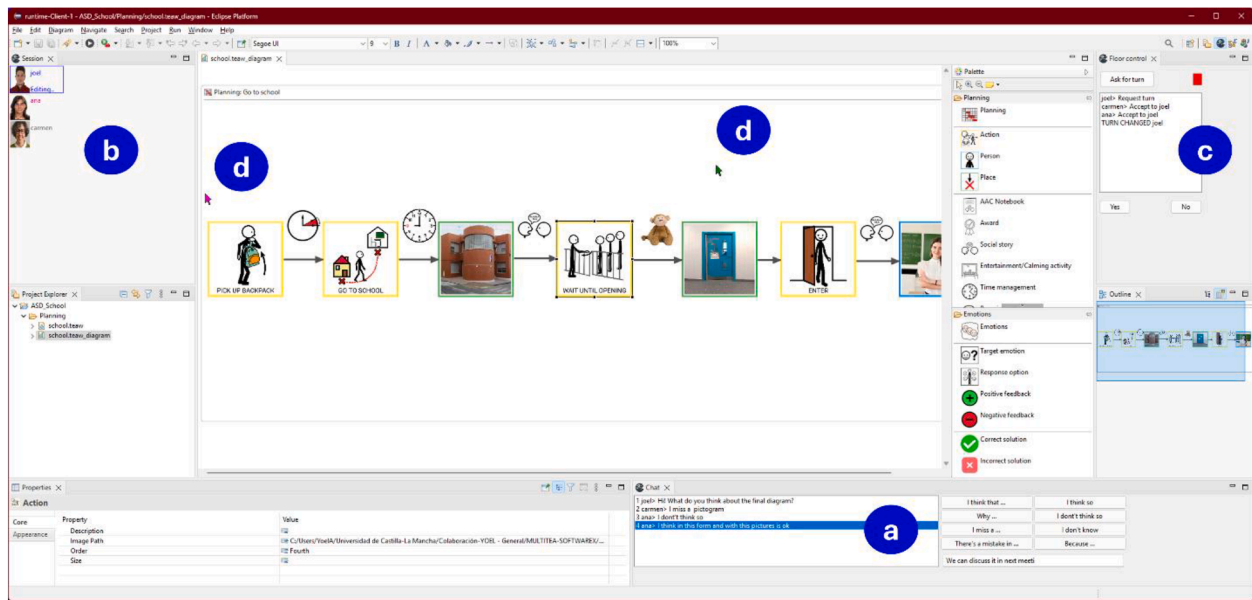


Fig. 4. Collaborative and awareness support mechanisms integrated into the collaborative graphical editor: (a) chat, (b) session panel, (c) turn-taking panel (semaphores) and (d) telepointers.

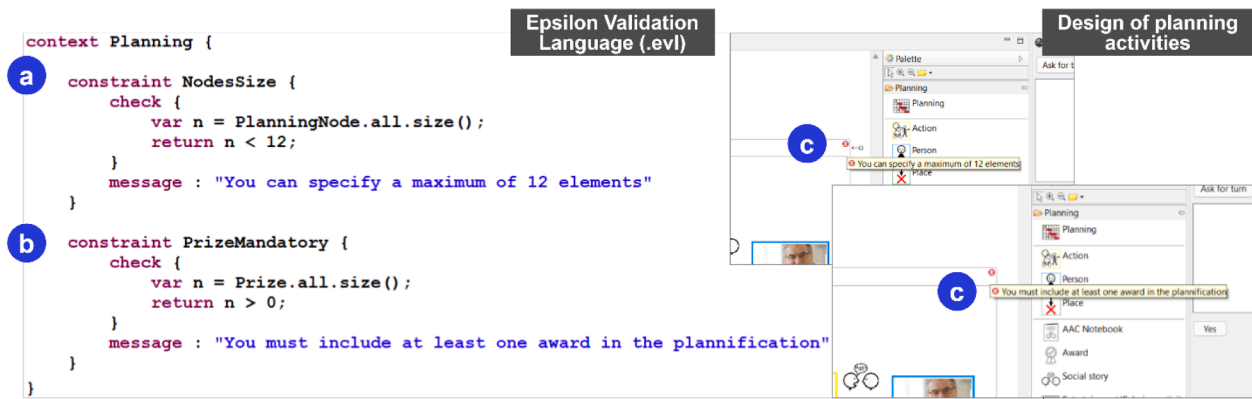


Fig. 5. Support for validating and correcting ASD activity designs: (a) specification in EVL of the restriction related to the number of steps in a plan, (b) specification in EVL of a UX guideline for applications aimed at users with ASD, and (c) notifications in the graphic editor about any errors or warnings detected in the diagram.

guideline-compliant solutions.

Thus, using the features supported by this collaborative component of CoGenASD, during a co-design session, all participants can jointly engage in the specification, customization and validation of the ASD activity specification and initiate the automatic generation of cross-platform instances of ASD application. Although this collaborative mode is entirely optional, it significantly enhances participation and personalization during the design of the ASD application.

2.3. Export and deployment component

Finally, the **export and deployment component** (Fig. 1c) converts the designed model into a deployable cross-platform web ASD application, generating application code, assets and configuration files for the multi-platform deployment of activities.

This component encapsulates the technological foundation and structural components required for deploying interactive, web-based tools capable of running on mobiles, desktops, tablets and large-format touch displays. Being web-based, the generated applications can be accessed on any device with a modern web browser, providing inherent cross-platform compatibility without additional adaptations.

This approach ensures that users can interact with the application across different ecosystems, including iOS, Android, Windows, and Linux, while maintaining a consistent user experience.

At its core, the generated application consists of an *index.html* file serving as the main entry point; a suite of JavaScript files (*teaw.js*, *gojs.js*) responsible for logic and interactivity; multiple CSS style sheets (*teaw-desktop.css*, *teaw-large.css*, *teaw-tablet.css* and *teaw-phone.css*) to ensure responsive design across various screen sizes; and an image repository containing the graphical assets for the GUI. The application logic is powered by GoJS,⁷ a JavaScript library that enables dynamic and customizable rendering and manipulation of diagrams.

3. Illustrative examples

This section describes a usage scenario of the proposed platform. Consider the following situation: “Two children—Emma and Ethan—classmates enrolled in the same school, attend a specialized care center for children with ASD during the summer holidays. With the beginning of the

⁷ <https://gojs.net/latest/>

school year approaching, educators, therapists, and families decide to use the CoGenASD application to anticipate and prepare the children for this new stage”

A few days before the beginning of the academic year, professionals and family members participate in co-design sessions aimed at defining and modelling a planning activity with CoGenASD. During these sessions, and thanks to the advantages of MDD, a visual sequence of anticipated steps is quickly constructed and fully understandable by both professionals and family members, who can incorporate pictograms and/or real images that represent the actions, locations, and individuals involved in the forthcoming scenario. The planning activity is subsequently enhanced with contextual annotations, including temporal cues, social stories, calming objects, and motivational incentives. The distributed collaborative editor implementing MDD enables therapists and parents (who may be in different locations, such as on vacation) to jointly validate the planning activity (Fig. 6).

Once the collaborative design phase is completed, CoGenASD automatically generates multiple views of the activity, which are deployed in two formats: a version for a multitouch screen installed in the specialized care center classroom that the children attend during the summer (Fig. 7a), and a mobile version for use by their families (Fig. 7b and c).

In the classroom, the multitouch screen supports an interactive group experience where the children can explore and become familiar with the anticipated steps and scenarios of their first day at school, several days in advance. This view includes a visual sequence combining pictograms and real images (e.g., photographs of the school building, the teacher, etc.), describing the entire process—from leaving home to being welcomed by the teacher—enabling tactile interaction with the anticipated routine. The multitouch interface fosters collaborative learning and shared engagement, while educators can present and explain the content collectively. Meanwhile, the mobile version allows families to accompany and monitor the child’s individual progress on the actual school day (Fig. 7d). Through their mobile devices, parents can also access and rehearse the routine in advance, reinforcing the steps at home and supporting the child’s preparation for the new situation.

In this way, the collaborative approach facilitated by CoGenASD between therapists and families contributes to a smoother adaptation of Emma and Ethan to the beginning of the school year. Furthermore, the dual interaction mode—in both classroom and home settings—enhances continuity and consistency in anticipation and planning, thereby reducing potential anxiety and promoting autonomy in children with ASD.

4. Impact

CoGenASD provides an **innovative solution** to a major challenge in the development of digital tools for individuals with ASD: the lack of accessible, customizable and inclusive software creation processes. This system is especially useful for designing applications for children. By combining co-design principles with a MDD approach, CoGenASD empowers multidisciplinary agents (including therapists and families or caregivers) to actively participate in the semi-automatic generation of cross-platform applications tailored to specific user profiles, contexts and goals. The tool facilitates the creation of *highly personalized activities*, allowing practitioners to define content types and interaction modalities that match the specific support requirements of each ASD individual. The integration of co-design techniques throughout the modelling

process also *enhances stakeholder engagement* and increased the relevance and *acceptance* of the resulting applications.

This model-driven approach offers significant **advantages** that *enhance* both the *development process* and the *quality* of the resulting therapeutic tools. *Efficiency* is achieved by automating the generation of key components of the ASD application, including its structure, appearance and behavioral constraints, thus minimizing manual coding efforts. *Reusability and consistency* are maintained through metamodels and generation templates, which can be adapted across different projects or deployment targets while preserving a standardized design philosophy. Built-in validation via EVL ensures that each ASD diagram meets structural, usability and therapeutic requirements before any application is generated, reinforcing both *reliability* and *quality*. Furthermore, the *scalability* of the model-driven infrastructure allows for the easy extension of the system to support new activity types, user interfaces or target platforms with minimal changes to the core models. This *flexibility* makes it possible to adapt the framework for diverse user needs, intervention goals and technological environments.

The CoGenASD framework exemplifies the *benefits of model-driven engineering in therapeutic software development*. This approach *can reduce time, cost and technical barriers*, ensuring tools are scalable, adaptable and aligned with best practices for interventions and user experience (UX) for ASD users. Beyond immediate benefits for researchers, practitioners and institutions, CoGenASD provides a *reproducible model* that can be extended to other populations with special needs, *accelerating the development of assistive technologies* and enabling new research avenues in technology-mediated interventions for neurodiverse individuals.

5. Limitations

Despite its merits, CoGenASD presents several limitations that should be acknowledged. First, the flexibility and expressiveness of the generated applications are currently constrained by the supported activity formats and types; future work aims to extend this by creating a taxonomy of activity types, incorporating multiple formats (pictograms, text, or combinations), and embedding multimedia resources. Second, the tool requires some level of training for stakeholders, which can be partially addressed through manuals, video tutorials, and guided support during the co-design process. Third, integration with external systems, such as AAC tools, and additional coding mechanisms (e.g., color-coding pictogram categories) are not yet fully implemented. Finally, the system’s handling of changes and unexpected events in activity plans, as well as its current support for progress tracking and gamification, remain areas for improvement.

Addressing these limitations is planned as future work, alongside the enhancement of collaborative communication channels and the expansion of empirical validation. While the system has been deployed and evaluated informally in a real setting in which a 6-year-old girl and an 8-year-old boy, their respective mothers and a psychologist to represent the use case in Section 3, a more comprehensive study with a larger and more diverse group of stakeholders is planned to fully assess the framework’s effectiveness and usability. As this evaluation involves sensitive data, the complete study design has already been submitted for approval to our university’s Ethics Committee, and authorization is currently pending.

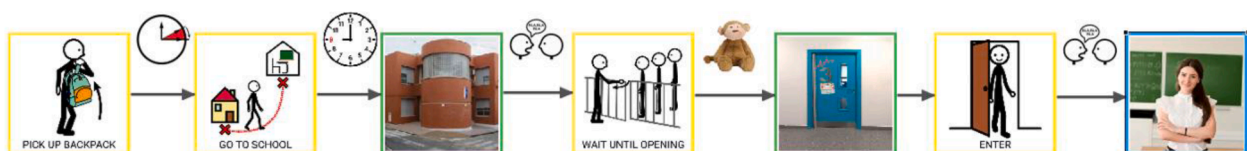


Fig. 6. Planning activity resulting from the individual design and co-design phases supported by CoGenASD editors.

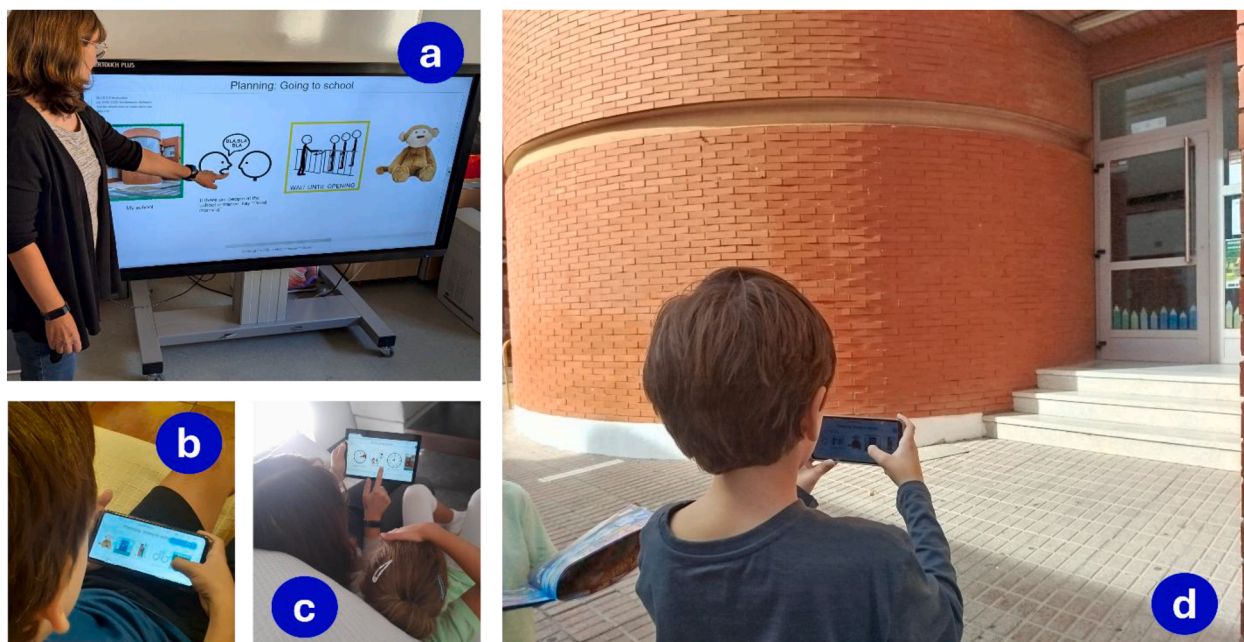


Fig. 7. Activity deployed across multiple interaction devices and settings: (a) at the specialized care center in the final days of summer (e.g., projecting onto an interactive whiteboard); at home, to anticipate the activity, (b) alone (e.g. using a smartphone) or (c) with the help of a family member (e.g. using a tablet), and (d) during its execution (e.g. using a smartphone).

6. Conclusions

CoGenASD represents a significant step forward in the development of inclusive digital tools for individuals with ASD. By integrating co-design methodologies with a model-driven development approach, the platform addresses the need for more accessible, participatory and scalable software solutions in this field. The tool enables the semi-automatic generation of cross-platform applications, fostering the active involvement of therapists, educators, families and other stakeholders throughout the development process.

The combination of personalization capabilities and participatory design contributes not only to improving the usability and acceptance of the resulting applications but also to reducing the technical and temporal barriers typically associated with the development of assistive technology.

Future work will focus on expanding the variety of activities supported. We also plan to integrate features such as activity restart, progress tracking, embedded multimedia and gamification elements. Enhancements in co-design support—such as richer communication channels—will foster more effective collaboration. Once these improvements are in place, broader validation studies involving ASD users and their families will be conducted to assess the framework's effectiveness and generalizability.

Moreover, CoGenASD lays the foundation for new research directions in the areas of collaborative modelling, inclusive technology design and technology-mediated intervention for neurodiverse populations. Its reproducible and extensible framework can be adapted to other collectives with special needs, supporting both scientific inquiry and real-world application. Ultimately, CoGenASD aims to contribute meaningfully to the inclusion, autonomy and well-being of individuals with ASD through accessible and adaptive digital innovation.

CRediT authorship contribution statement

Yoel Arroyo: Writing – original draft, Software, Methodology, Conceptualization. **Ana I. Molina:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Carmen Lacave:** Writing – review & editing, Supervision. **Miguel Á. Redondo:**

Funding acquisition.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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